HardScope Lead Analyst Challenge

GlowMetric Q1 2025 — Submission Summary

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1. The Challenge

HardScope builds measurement systems, attribution logic, and reporting frameworks that turn creator content into deterministic brand ROI. The challenge: act as lead analyst on a creator program and answer three questions for the partnerships team:

- **What did we get from this creator program?**
- **Which creators, platforms, and content formats are working?**
- **Where should we reallocate budget next quarter?**

The output required: real data, a defensible measurement framework, an analytics-ready dataset, a dashboard, and an executive readout, all within 48 hours.

2. Approach

Brand context: GlowMetric, a fictional DTC skincare brand launching an SPF serum line, activating 10 beauty/skincare creators across YouTube and TikTok for Q1 2025.

Data

- YouTube: Real data via YouTube Data API v3 - 138 videos across 10 creators, pulled live
- TikTok: Benchmark-grounded synthetic data (TikTok Research API requires 4–6 week approval). Benchmarks sourced from CreatorIQ 2024, Influencer Marketing Hub 2024, Social Insider Q4 2024
- Combined: 288-row unified analytics table - one row per creator per video, normalized across both platforms

Pipeline

- Python (pandas) for data ingestion, normalization, and feature engineering
- SQLite for 8 SQL views: creator scorecard, funnel analysis, tier analysis, format performance, incrementality proxy, alerting
- Power BI for 3-page dashboard with DAX measures, conditional formatting, and platform slicers

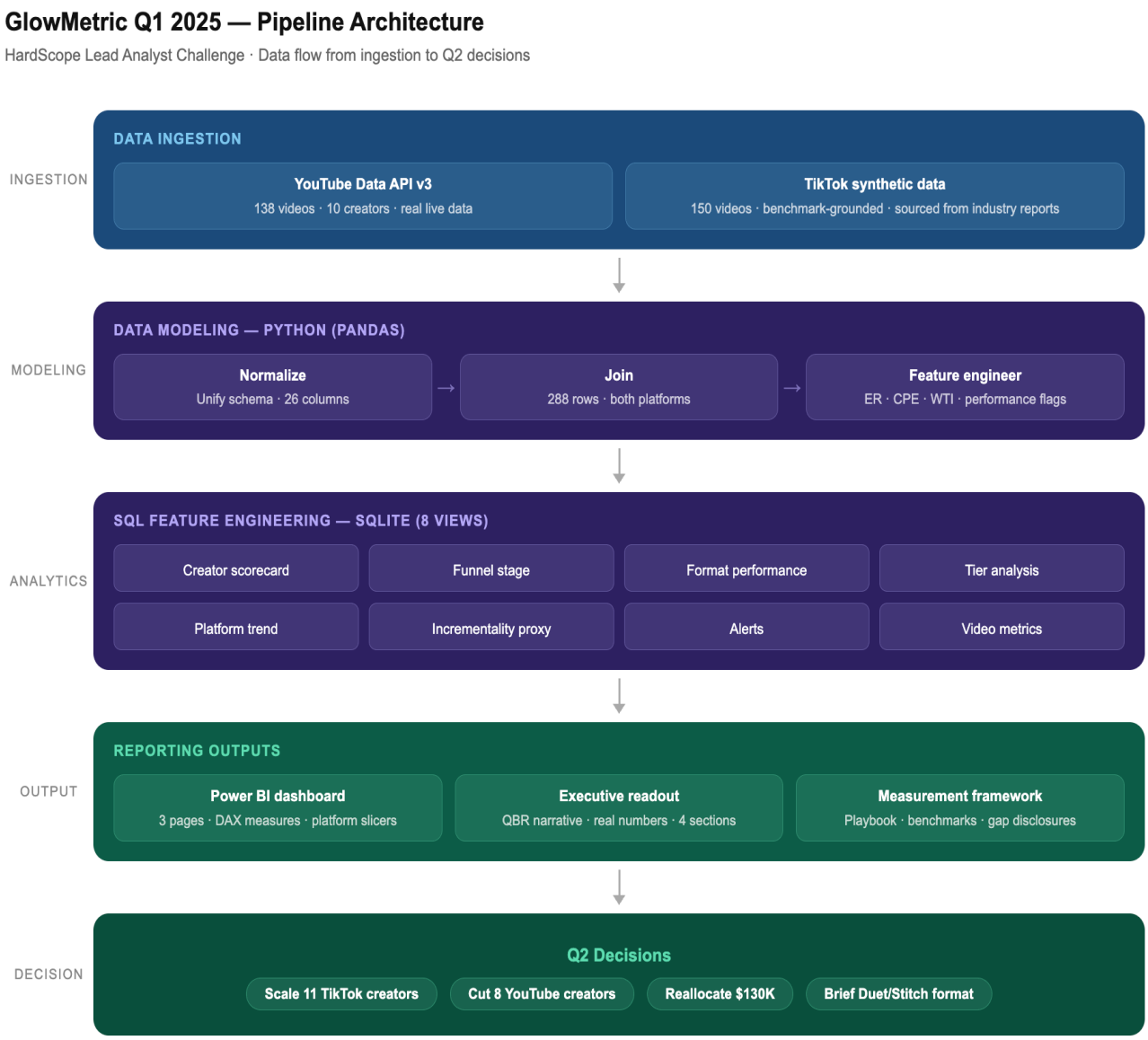
Framework

Three-tier measurement funnel covering Awareness, Engagement, and Consideration — with defined benchmarks, lift definitions, cost efficiency metrics, alerting logic, and attribution gap disclosures. Designed as a reusable playbook.

3. Requirements vs. Deliverables

Requirement	What We Built	Status
Real data from a legitimate source	YouTube Data API v3 - 138 real videos, 10 creators	Done
Show how you got it (code / API calls)	youtube_pull_v2.py - full API script with dynamic channel resolution	Done
A measurement framework (not just metrics)	3-tier funnel: Awareness → Engagement → Consideration with benchmarks, lift definitions, gap disclosures	Done
Modeled dataset ready for analytics	288-row unified analytics table, 26 cols, one row per video per platform	Done
Joins, transformations, feature engineering	Python data_model.py + 8 SQL views via SQLite	Done

Dashboard a partnerships lead would use	3-page Power BI: Program Overview, Creator Scorecard, Funnel & Formats	Done
Summary stats + funnel view + sortable table	5 KPI cards, platform bars, creator table with ER color scale + Scale/Cut badges	Done
Executive readout (1–2 pages)	4-section QBR narrative with real numbers, headline findings, Q2 recommendations	Done
README a teammate follows in 5 min	Full README: run steps, API key instructions, framework summary, architecture decisions	Done



4. Bonus Signal - Completed

Bonus Signal	What Was Achieved
Multiple data sources normalized into single schema	YouTube (API) + TikTok (synthetic) joined into one 26-col analytics table with unified field names
Basic incrementality logic	Split the campaign into three phases (before, during, and late) and compared engagement rates across each phase to estimate whether performance improved over time
Thoughtful benchmarks	Used real industry reports to define what "good" looks like for beauty creators, every video gets automatically labeled as Strong, Average, or Weak based on those standards
Simple automation concept	run_sql.py acts as the pipeline runner. The pipeline runs in one command. The README also outlines how this could be set up to refresh automatically every week without manual work
Alerting / flagging logic	Any creator whose engagement drops more than 20% compared to the previous month gets automatically flagged and every creator gets a Scale, Optimize, or Cut label based on their overall performance
Measurement playbook template	The entire framework is built so it can be reused for a different brand in under 2 hours just swap the brand name, update the category benchmarks, and it's ready to go

5. Headline Findings

- 239.4M total views and 30.3M engagements across 288 videos, 10 creators, 2 platforms
- TikTok generated 87% of all engagements at 22.6× lower cost per engagement than YouTube ($0.02 vs $0.55)
- Nano creators outperform Mega by 2.4× on ER - View/Follower ratio exceeds 1.0, meaning content reaches beyond the existing follower base
- Duet/Stitch format: 92% Strong ER rate - most consistent format in the program, currently underused
- 8 YouTube creators flagged for Cut, combined modeled spend $130K with negligible engagement return
- Q2 recommendation: Reallocate $130K from Cut YouTube creators to TikTok Nano/Micro, estimated 13M additional engagements at current CPE

6. With Another Week

- **Build an agentic scraper** for TikTok: instead of waiting months for API approval, use an AI agent with browser automation to scrape public TikTok profile pages (views, likes, comments, captions) for the specific creators in the program. Scrape on a schedule, parse with an LLM for caption themes and product mentions and feed it into the same pipeline. Faster than any API and richer than raw metrics alone.
- **Stitch YouTube and TikTok posting schedules together**: right now, the two platforms are analyzed independently. A creator who posts a YouTube tutorial on Monday and a TikTok GRWM on Wednesday about the same product might be driving compounding lift. Cross-platform sequencing analysis: does a YouTube post followed by TikTok within 7 days lift engagement on both? is something I'd want to model.
- **Wire in Google Trends as a live consideration signal**: pull branded search volume for "GlowMetric serum" weekly using the pytrends library, overlay it with creator posting dates, and build a simple lag-correlation model. Not a causal claim, but a directional signal that tells you whether the content is moving brand interest or just platform metrics.
- **Incrementality holdout**: randomly pause 2 creators for 4 weeks in Q2. Compare branded search volume vs. active creators to estimate true incremental lift.
- **Audience demographic validation**: run top 5 Scale creators through CreatorIQ or StatSocial API to confirm 18–34 beauty buyer alignment before increasing budgets.
- **Productionize the pipeline**: Apache Airflow DAG with weekly scheduled refresh, Slack alerting when any creator's ER drops >20%, and a parameterized playbook template deployable for a new brand in under 2 hours.